

Homework 3

① Find the sum for each of the following series:

$$\begin{aligned} a) \sum_{n \geq 2} \ln \left(1 - \frac{1}{n^2} \right) &= \sum_{n \geq 2} \ln \frac{n^2 - 1}{n^2} = \sum_{n \geq 2} \ln \frac{(n-1)(n+1)}{n^2} = \\ &= \sum_{n \geq 2} \ln (n-1)(n+1) - \ln n^2 = \sum_{n \geq 2} \ln (n-1)(n+1) - 2 \ln n = \\ &= \sum_{n \geq 2} \ln (n-1)(n+1) - \ln n - \ln n = \\ &= \sum_{n \geq 2} \ln (n-1) - \ln n + \ln (n+1) - \ln n \end{aligned}$$

Let $a_n = \ln n - \ln (n+1)$

$\Rightarrow a_{n-1} = \ln (n-1) - \ln n$

$$\begin{aligned} a_{n-1} - a_n &= [\ln (n-1) - \ln n] - [\ln n - \ln (n+1)] = \\ &= \ln (n-1) - \ln n - \ln n + \ln (n+1), \text{ which is the} \\ &\quad \text{term of our sum} \end{aligned}$$

$\Rightarrow \ln \left(1 - \frac{1}{n^2} \right) = a_{n-1} - a_n$

$$\begin{aligned} S_N &= \sum_{n=2}^N (a_{n-1} - a_n) = a_1 - \cancel{a_2} + \cancel{a_2} - \cancel{a_3} + \dots + \cancel{a_{N-1}} - a_N = \\ &= a_1 - a_N = (\ln 1 - \ln 2) - (\ln N - \ln (N+1)) = \\ &= -\ln 2 - \ln N + \ln (N+1) = \\ &= -\ln 2 + \ln \frac{N+1}{N} \end{aligned}$$

$$\begin{aligned} \lim_{N \rightarrow \infty} S_N &= \lim_{N \rightarrow \infty} \left(-\ln 2 + \ln \frac{N+1}{N} \right) = \\ &= -\ln 2 + \lim_{N \rightarrow \infty} \ln \frac{N+1}{N} = -\ln 2 + \ln \left(\lim_{N \rightarrow \infty} \frac{N+1}{N} \right) = \\ &= -\ln 2 + \ln 1 = -\ln 2 \end{aligned}$$

$\Rightarrow \sum_{n \geq 2} \ln \left(1 - \frac{1}{n^2} \right) = -\ln 2$

$$b) \sum_{n \geq 1} \frac{n+1}{3^n} = \sum_{n \geq 1} \frac{n}{3^n} + \sum_{n \geq 1} \frac{1}{3^n} \quad (1)$$

$$\begin{aligned} \bullet \sum_{n \geq 1} \frac{1}{3^n} &= \sum_{n \geq 1} \left(\frac{1}{3}\right)^n = \left(\sum_{n \geq 0} \left(\frac{1}{3}\right)^n\right) - \left(\frac{1}{3}\right)^0 = \\ &= \frac{1}{1 - \frac{1}{3}} - 1 = \frac{1}{\frac{2}{3}} - 1 = \frac{3}{2} - 1 = \frac{1}{2} \quad (2) \end{aligned}$$

• By differentiating the geometric series, we get that

$$\sum_{n \geq 1} n x^n = \frac{x}{(1-x)^2}$$

$$\sum_{n \geq 1} \frac{n}{3^n} = \sum_{n \geq 1} n \cdot \frac{1}{3^n} = \sum_{n \geq 1} n \cdot \left(\frac{1}{3}\right)^n$$

In this case $x = \frac{1}{3}$

$$\Rightarrow \sum_{n \geq 1} \frac{n}{3^n} = \frac{\frac{1}{3}}{\left(\frac{1}{3} - \frac{1}{3}\right)^2} = \frac{\frac{1}{3}}{\left(\frac{2}{3}\right)^2} = \frac{\frac{1}{3}}{\frac{4}{9}} = \frac{1}{3} \cdot \frac{9}{4} = \frac{3}{4} \quad (3)$$

$$\xRightarrow{(1)(2)(3)} \sum_{n \geq 1} \frac{n+1}{3^n} = \frac{3}{4} + \frac{1}{2} = \frac{5}{4}$$

$$c) \sum_{n \geq 1} \frac{n}{n^4 + n^2 + 1}$$

$$n^4 + n^2 + 1 = (n^2)^2 + 2n^2 + 1 - n^2 = (n^2 + 1)^2 - n^2 = (n^2 - n + 1)(n^2 + n + 1)$$

$$\frac{n}{n^4 + n^2 + 1} = \frac{1}{2} \cdot \left(\frac{1}{n^2 - n + 1} - \frac{1}{n^2 + n + 1} \right)$$

$$\text{Let } a_n = \frac{1}{n^2 - n + 1}$$

$$a_{n+1} = \frac{1}{(n+1)^2 - (n+1) + 1} = \frac{1}{n^2 + 2n + 1 - n - 1 + 1} = \frac{1}{n^2 + n + 1}$$

$$a_n - a_{n+1} = \frac{1}{n^2 - n + 1} - \frac{1}{n^2 + n + 1} \Rightarrow \text{the term of our sum is } \frac{1}{2}(a_n - a_{n+1})$$

$$S_N = \sum_{m=1}^N \frac{1}{2} (a_m - a_{m+1}) = \frac{1}{2} \sum_{m=1}^N a_m - a_{m+1} =$$

$$= \frac{1}{2} (\cancel{a_1} - \cancel{a_2} + \cancel{a_2} - \cancel{a_3} + \dots + \cancel{a_{N-1}} - a_N) = \frac{N^2 - N + 1}{2}$$

$$= \frac{1}{2} (a_1 - a_N) = \frac{1}{2} \left(\frac{1}{1^2 - 1 + 1} - \frac{1}{N^2 - N + 1} \right) = \frac{1}{2} \left(1 - \frac{1}{N^2 - N + 1} \right) =$$

$$= \frac{1}{2} \cdot \frac{N^2 - N + 1 - 1}{N^2 - N + 1} = \frac{1}{2} \cdot \frac{N^2 - N}{N^2 - N + 1}$$

$$\lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \frac{1}{2} \cdot \frac{N^2 - N}{N^2 - N + 1} = \frac{1}{2} \cdot \lim_{N \rightarrow \infty} \frac{N^2 - N}{N^2 - N + 1} = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

② Study the convergence of the following series:

a) $\sum_{m=1}^{\infty} \underbrace{\frac{x^m}{m^p}}_{a_m}, x > 0, p \in \mathbb{N}$

Ratio Test: $\lim_{m \rightarrow \infty} \frac{a_{m+1}}{a_m} = \lim_{m \rightarrow \infty} \frac{\frac{x^{m+1}}{(m+1)^p}}{\frac{x^m}{m^p}} = \lim_{m \rightarrow \infty} \frac{x^{m+1}}{(m+1)^p} \cdot \frac{m^p}{x^m} =$

$$= \lim_{m \rightarrow \infty} x \cdot \left(\frac{m}{m+1} \right)^p = x$$

Conclusion: $\begin{cases} x > 1 \Rightarrow \text{the series diverges} \\ 0 < x < 1 \Rightarrow \text{the series converges} \\ x = 1 \Rightarrow \sum_{m=1}^{\infty} \frac{1}{m^p} \begin{cases} p > 1 \Rightarrow \text{the series converges} \\ p \leq 1 \Rightarrow \text{the series diverges} \end{cases} \end{cases}$

b) $\sum_{m=2}^{\infty} \frac{1}{(\ln m)^{\ln m}} = \sum_{m=2}^{\infty} \frac{1}{e^{\ln((\ln m)^{\ln m})}} =$

$$= \sum_{m=2}^{\infty} \frac{1}{e^{(\ln m) \cdot \ln(\ln m)}} = \sum_{m=2}^{\infty} \frac{1}{(\underbrace{e^{\ln m}}_m)^{\ln(\ln m)}} = \sum_{m=2}^{\infty} \frac{1}{\underbrace{m^{\ln(\ln m)}}_{=x_m}}$$

Compare the series with $\sum_{m=2}^{\infty} \underbrace{\frac{1}{m^2}}_{=y_m}$

$$\lim_{m \rightarrow \infty} \frac{x_m}{y_m} = \lim_{m \rightarrow \infty} \frac{\frac{1}{m^{\ln(\ln m)}}}{\frac{1}{m^2}} = \lim_{m \rightarrow \infty} \frac{m^2}{m^{\ln(\ln m)}} \rightarrow \infty = 0$$

$\sum_{m=2}^{\infty} \frac{1}{m^2}$ is convergent $\Rightarrow \sum_{m=2}^{\infty} \frac{1}{(\ln m)^{\ln m}}$ is also convergent

$$c) \sum_{n \geq 1} (\sqrt[n]{n} - 1) = \sum_{n \geq 1} \underbrace{\left(n^{\frac{1}{n}} - 1 \right)}_{e^{\frac{\ln n}{n}} - 1} = \sum_{n \geq 1} \underbrace{\left(e^{\frac{\ln n}{n}} - 1 \right)}_{= x_n}$$

Compare with $\sum_{n \geq 1} \underbrace{\frac{\ln n}{n}}_{= y_n}$

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{e^{\frac{\ln n}{n}} - 1}{\frac{\ln n}{n}}$$

take $y = \frac{\ln n}{n}$, which goes to 0 as $n \rightarrow \infty$

$$\lim_{y \rightarrow 0} \frac{e^y - 1}{y} = 1 \Rightarrow \sum_{n \geq 1} (\sqrt[n]{n} - 1) \text{ has the same nature as } \sum_{n \geq 1} \frac{\ln n}{n}$$

Find the nature of $\sum_{n \geq 1} \underbrace{\frac{\ln n}{n}}_{= x_n}$

Compare with the harmonic series, $\sum_{n \geq 1} \underbrace{\frac{1}{n}}_{= y_n}$

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{\frac{\ln n}{n}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \ln n = \infty$$

$\sum_{n \geq 1} \frac{1}{n}$ is divergent $\Rightarrow \sum_{n \geq 1} \frac{\ln n}{n}$ is also divergent

We proved that $\sum_{n \geq 1} (\sqrt[n]{n} - 1)$ has the same nature

as $\sum_{n \geq 1} \frac{1}{n} \Rightarrow \sum_{n \geq 1} (\sqrt[n]{n} - 1)$ is also divergent

$$d) \sum_{n \geq 1} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n} \cdot \frac{1}{2n+1} = a_n$$

~~the series~~ Raabe Duhamel Test

$$R = \lim_{n \rightarrow \infty} n \left(\frac{a_n}{a_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n \cdot (2n+1)} \cdot \frac{2 \cdot 4 \cdot \dots \cdot 2n \cdot (2n+2) \cdot (2n+3)}{1 \cdot 3 \cdot \dots \cdot (2n+1) \cdot (2n+1)} - 1 \right)$$

$$= \lim_{n \rightarrow \infty} n \cdot \left(\frac{(2n+2)(2n+3)}{(2n+1)^2} - 1 \right) = \lim_{n \rightarrow \infty} n \cdot \frac{4n^2 + 10n + 6 - 4n^2 - 4n - 1}{4n^2 + 4n + 1} =$$

$$= \lim_{n \rightarrow \infty} n \cdot \frac{6n + 5}{4n^2 + 4n + 1} = \lim_{n \rightarrow \infty} \frac{6n^2 + 5n}{4n^2 + 4n + 1} = \frac{6}{4} = \frac{3}{2} > 1 \Rightarrow$$

$\Rightarrow$ convergence

$$\Rightarrow \sum_{n \geq 1} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n} \cdot \frac{1}{2n+1} \text{ is convergent}$$

③ Start with an equilateral triangle of side 1. For each side, remove the middle third and add there another equilateral triangle. Repeat this process at each iteration.

- How many sides are there at iteration n ?
- What is the limit of the perimeter and the area?

• Iteration 0: equilateral triangle $\Rightarrow N_0 = 3$ sides ($3 \cdot 4^0$)

Iteration 1: each side of the figure is replaced by 4 smaller segments

— becomes 

$$\Rightarrow N_{01} = 3 \cdot 4 = 12 \text{ sides } (3 \cdot 4^1)$$

Iteration 2: each of 12 smaller sides is replaced by

$$4 \text{ segments } \Rightarrow N_2 = 12 \cdot 4 = 48 \text{ sides } (3 \cdot 4^2)$$

⋮

Iteration n : As a general rule, for iteration n , we have $N_n = 3 \cdot 4^n$ sides

- Perimeter at iteration n (as $n \rightarrow \infty$)

At the beginning: side length $\overset{(\Delta_0 = 1)}{=} 1$ $\Big| \Rightarrow P = 1 \cdot 3 = 3$
3 sides ($P_0 = 3$)

With each iteration, each side is divided into 4 smaller segments $\Rightarrow$ the new side length $= \frac{1}{3}$ of the previous side length

$$\Rightarrow \Delta_n = \left(\frac{1}{3}\right)^n$$

$$P_n = N_n \cdot \Delta_n = \underset{\substack{\text{as} \\ \text{computed} \\ \text{previously} \\ (3 \cdot 4^n)}}{\uparrow} 3 \cdot 4^n \cdot \left(\frac{1}{3}\right)^n = 3 \cdot 4^n \cdot \frac{1}{3^n} = 3 \cdot \frac{4^n}{3^n} = 3 \cdot \left(\frac{4}{3}\right)^n$$

$$n \rightarrow \infty \Rightarrow \lim_{n \rightarrow \infty} P_n = \lim_{n \rightarrow \infty} 3 \cdot \left(\frac{4}{3}\right)^n = 3 \cdot \lim_{n \rightarrow \infty} \left(\frac{4}{3}\right)^n = 3 \cdot \infty = \infty$$

Area at iteration n (as $n \rightarrow \infty$)

At the beginning: side length = 1 ($s_0 = 1$)
(Iteration 0) $\Rightarrow A_0 = \frac{s_0^2 \sqrt{3}}{4} = \frac{\sqrt{3}}{4}$

At Iteration 1: Each of the 3 sides gets a new equilateral triangle of side length ($s_1 = \frac{1}{3}$) added

The area of a such triangle is

$$\frac{s_1^2 \sqrt{3}}{4} = \frac{\left(\frac{1}{3}\right)^2 \sqrt{3}}{4} = \frac{\frac{\sqrt{3}}{9}}{4} = \frac{\sqrt{3}}{36}$$

But we add 3 such triangles $\Rightarrow 3 \cdot \frac{\sqrt{3}}{36} = \frac{\sqrt{3}}{12}$
 $\Rightarrow A_1 = A_0 + \frac{\sqrt{3}}{12}$ (3 $\cdot$ 4¹)

At Iteration 2: As previously shown, we now have 12 sides
Each side gets a new equilateral triangle of side length $s_2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$

The area of a such triangle is

$$\frac{s_2^2 \sqrt{3}}{4} = \frac{\left(\frac{1}{9}\right)^2 \sqrt{3}}{4} = \frac{\frac{1}{81} \sqrt{3}}{4} = \frac{\sqrt{3}}{324}$$

But we add 12 such triangles $\Rightarrow 12 \cdot \frac{\sqrt{3}}{324} = \frac{\sqrt{3}}{27}$
 $\Rightarrow A_2 = A_1 + \frac{\sqrt{3}}{27}$

...

At Iteration n : We have $N_{n-1} = 3 \cdot 4^{n-1}$ new triangles that are added, with a side length $s_n = \left(\frac{1}{3}\right)^n$

$\Rightarrow$ at this step, the added area is

$$\begin{aligned} 3 \cdot 4^{n-1} \cdot \frac{\left[\left(\frac{1}{3}\right)^n\right]^2 \sqrt{3}}{4} &= \frac{3 \cdot 4^n}{4} \cdot \frac{\frac{\sqrt{3}}{9^n}}{4} = \frac{3\sqrt{3} \cdot 4^n}{16 \cdot 9^n} = \\ &= \frac{3\sqrt{3} \cdot \left(\frac{4}{9}\right)^n}{16} = \frac{3\sqrt{3}}{16} \cdot \left(\frac{4}{9}\right)^n \end{aligned}$$

general
formula
for A_n

$$\text{Total area} = A_0 + \sum_{k=1}^{\infty} A_k = \frac{\sqrt{3}}{4} + \sum_{k=1}^{\infty} \frac{3\sqrt{3}}{16} \cdot \left(\frac{4}{9}\right)^k =$$

$$= \frac{\sqrt{3}}{4} + \frac{3\sqrt{3}}{16} \sum_{k=1}^{\infty} \left(\frac{4}{9}\right)^k$$

geometric series with $q = \frac{4}{9} (< 1)$

$$\Rightarrow \sum_{k=1}^{\infty} \left(\frac{4}{9}\right)^k = \frac{\frac{4}{9}}{1 - \frac{4}{9}} = \frac{\frac{4}{9}}{\frac{5}{9}} = \frac{4}{9} \cdot \frac{9}{5} = \frac{4}{5}$$

$$\text{Total Area} = \frac{\sqrt{3}}{4} + \frac{3\sqrt{3}}{16} \cdot \frac{4}{5} = \frac{\sqrt{3}}{4} + \frac{3\sqrt{3}}{20} = \frac{5\sqrt{3}}{20} + \frac{3\sqrt{3}}{20} = \frac{8\sqrt{3}}{20} = \frac{2\sqrt{3}}{5}$$

Conclusion: $\begin{cases} \text{perimeter: infinite} \\ \text{area: } \frac{2\sqrt{3}}{5} \text{ (finite)} \end{cases}$